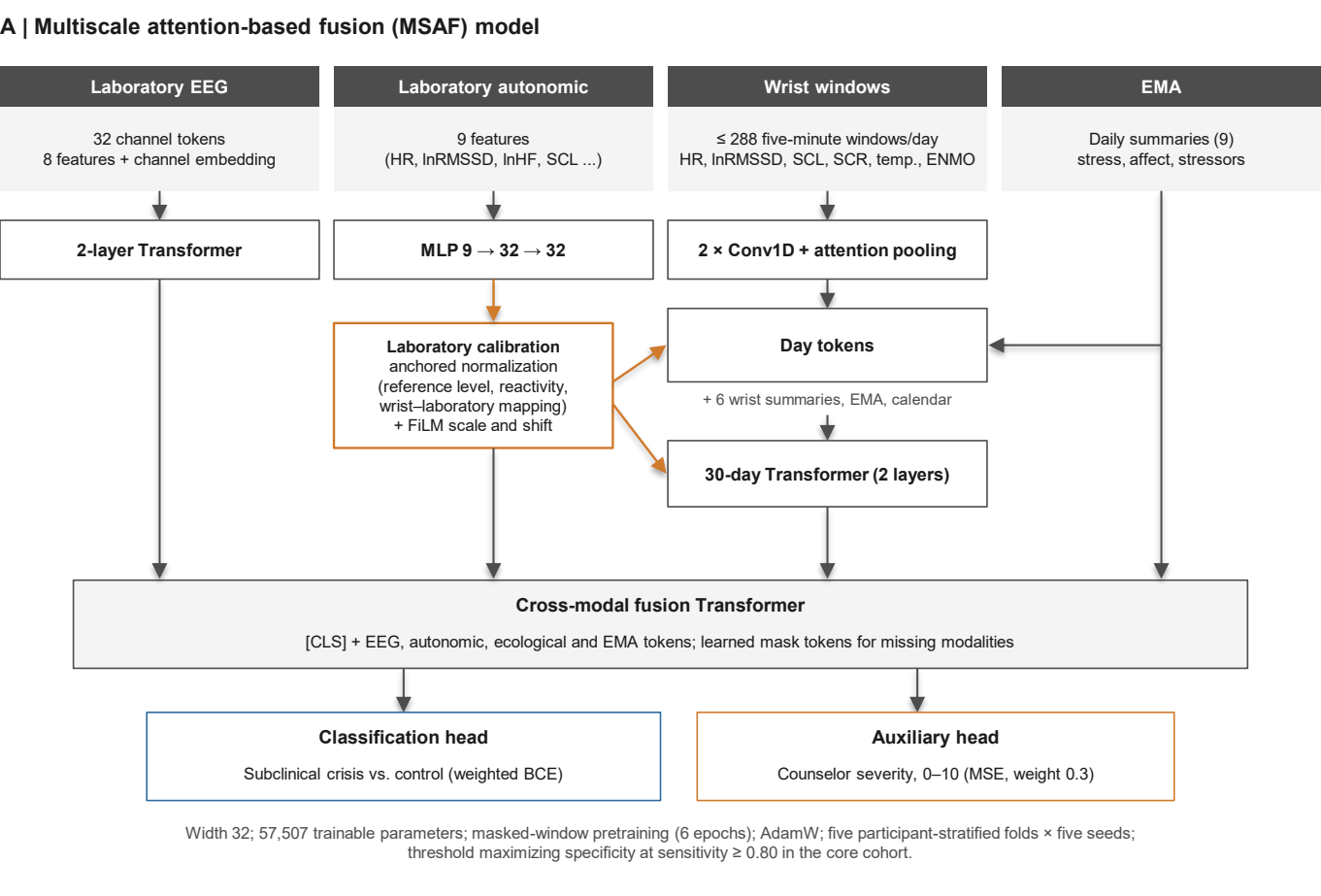

A | Multiscale attention-based fusion (MSAF) model
Laboratory EEG
32 channel tokens
8 features + channel embedding
Laboratory autonomic
9 features
(HR, lnRMSSD, lnHF, SCL ...)
Wrist windows
≤ 288 five-minute windows/day
HR, lnRMSSD, SCL, SCR, temp., ENMO
EMA
Daily summaries (9)
stress, affect, stressors
2-layer Transformer
MLP 9 → 32 → 32
2 × Conv1D + attention pooling
Laboratory calibration
anchored normalization
(reference level, reactivity,
wrist–laboratory mapping)
+ FiLM scale and shift
Day tokens
+ 6 wrist summaries, EMA, calendar
30-day Transformer (2 layers)
Cross-modal fusion Transformer
[CLS] + EEG, autonomic, ecological and EMA tokens; learned mask tokens for missing modalities
Classification head
Subclinical crisis vs. control (weighted BCE)
Auxiliary head
Counselor severity, 0–10 (MSE, weight 0.3)
Width 32; 57,507 trainable parameters; masked-window pretraining (6 epochs); AdamW; five participant-stratified folds × five seeds;
threshold maximizing specificity at sensitivity ≥ 0.80 in the core cohort.